Linear Number Patterns – Extra Practice

These questions are extra practice in the same style as Questions 1 and 2 of the *Linear number patterns* worksheet. For each question: (a) draw the next pattern, (b) complete the table, (c) find a formula, then (d)–(e) use your formula to answer questions about much larger patterns. Questions 1–3 are *linear* (add a constant amount each step); Question 4 is *geometric* (multiply by a constant amount each step).

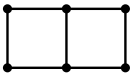
Question 1 – Matchstick Squares

This pattern shows the number of matchsticks needed to make a row of joined squares.

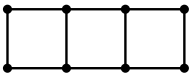
1:



2:



3:



(a) Draw the next pattern (4 squares in a row).

Draw your pattern here.

(b) Copy and complete the table.

S		1	2	3	4	5	12
M							

(c) Construct a formula for the number of matches (M) needed for any number of squares (S).

(d) If a row had 40 squares, how many matches would be needed?

(e) If a row had 75 squares, how many matches would be needed?

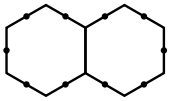
Question 2 – Hexagonal Picnic Tables

A campsite joins hexagonal picnic tables end to end in a row, as shown below (each dot is a seat). When two tables are pushed together, the shared side can no longer seat anyone.

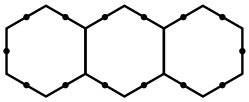
1:



2:



3:



(a) Draw the next pattern (4 hexagonal tables joined in a row).

Draw your pattern here.

(b) Copy and complete the table below.

T		1	2	3	4	5	25
C							

(c) Construct a formula for the number of seats (C) needed for any number of tables (T).

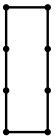
(d) If the campsite joined 40 tables in a row, how many seats would there be?

(e) If the campsite needed to seat 70 people in a single row of joined tables, how many tables would be needed?

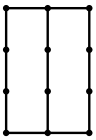
Question 3 – Matchstick Towers

This pattern shows the number of matchsticks needed to make a row of joined towers (each tower is 1 matchstick wide and 3 matchsticks tall). Neighbouring towers share a whole wall – 3 matchsticks' worth of overlap.

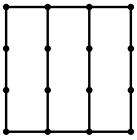
1:



2:



3:



(a) Draw the next pattern (4 towers in a row).

Draw your pattern here.

(b) Copy and complete the table.

T		1	2	3	4	5	20
M							

(c) Construct a formula for the number of matches (M) needed for any number of towers (T).

(d) If a row had 30 towers, how many matches would be needed?

(e) If a row had 60 towers, how many matches would be needed?

Question 4 – Paper Cutting (Geometric)

A square sheet of paper is cut with straight vertical cuts, each one cutting every piece made so far exactly in half. This pattern is *geometric*, not linear: the number of pieces is *multiplied* by the same amount at each stage.

1:



2:



3:



(a) Draw the next pattern (stage 4).

Draw your pattern here.

(b) Copy and complete the table below.

n	1	2	3	4	5	7
P						

(c) Construct a formula for the number of pieces (P) at stage n. (Hint: write a few terms as powers of the ratio you found above.)

(d) How many pieces are there at stage 8?

(e) How many pieces are there at stage 10?

Section 5 – Sequences in Words

Word problem 1

A candle is 24 cm tall when it is lit. Every hour it burns down, it loses 4 cm of height.

(a) Copy and complete the table below, using the words above.

h		1	2	3	4	5
H						

(b) Construct a formula for the height (H) after h hours.

(c) After how many hours does the candle burn out completely (height $= 0$)?

(d) Your formula gives a height for $h = 8$. Work it out, then explain why this answer doesn't make sense for the real candle.

Word problem 3

A water tank holds 500 litres when full. It is draining at a steady rate of 20 litres every minute.

(a) Copy and complete the table below, using the words above.

t		1	2	3	4	5
V						

(b) Construct a formula for the volume (V) remaining after t minutes.

(c) How many litres are left after 12 minutes?

(d) After how many minutes is the tank completely empty?

Word problem 2

A chess club has 15 members at the start of the year. Every week, 4 new people join the club (and nobody leaves).

(a) Copy and complete the table below, using the words above.

w		1	2	3	4	5	25
M							

(b) Construct a formula for the number of members (M) after w weeks.

(c) How many members would there be after 10 weeks?

(d) After how many weeks would the club have 99 members?

Word problem 4

Mia has \$80 saved already. Every week, she adds another \$12 to her savings.

(a) Copy and complete the table below, using the words above.

w		1	2	3	4	5
S						

(b) Construct a formula for her savings (S) after w weeks.

(c) How much would she have saved after 15 weeks?

(d) After how many weeks would she have saved \$272?

Section 6 – Fit the Numbers

For each scenario below, only two values are given – not the starting amount or the amount added each time. Work those out first, then fit the rest of the numbers to the table.

Scenario A

A tree is planted in a garden. It grows by the same amount every year. After 2 years it is 140 cm tall. After 6 years it is 220 cm tall.

(a) What is the common difference (the growth per year)?

(b) Copy and complete the table below (including year 0).

y		1	2	3	4	5	6
H			140				220

(c) Construct a formula for the height (H) after y years.

(d) How tall is the tree after 8 years?

(e) After how many years is the tree 320 cm tall?

Scenario B

A gym charges a joining fee plus the same amount for every week of membership. After 2 weeks, a member has paid \$115 in total. After 6 weeks, they have paid \$215 in total.

(a) What is the common difference (the cost per week)?

(b) Copy and complete the table below (including week 0).

w		1	2	3	4	5	6
C			115				215

(c) Construct a formula for the total cost (C) after w weeks.

(d) What would the total cost be after 10 weeks?

(e) After how many weeks would the total cost reach \$440?

Scenario C

A tank is being filled with water at a steady rate. After 2 minutes it holds 28 litres. After 6 minutes it holds 76 litres.

(a) What is the common difference (the fill rate per minute)?

(b) Copy and complete the table below (including minute 0).

t		1	2	3	4	5	6
V			28				76

(c) Construct a formula for the volume (V) after t minutes.

(d) How many litres are in the tank after 10 minutes?

(e) After how many minutes does the tank hold 244 litres?

Scenario D

A candle burns down at a steady rate. After 2 hours it is 30 cm tall. After 6 hours it is 10 cm tall.

(a) What is the common difference (the height lost per hour)?

(b) Copy and complete the table below (including hour 0).

h		1	2	3	4	5	6
H			30				10

(c) Construct a formula for the height (H) after h hours.

(d) How tall is the candle after 7 hours?

(e) After how many hours has the candle completely burnt out?